MIL-C-44161A
9 December 1987
SUPERSEDING
MIL-C-44161
28 February 1985

MILITARY SPECIFICATION

CLOTH, KNITTED, TERRY, SYNTHETIC FIBERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers polypropylene and polyester knitted terry cloth.

1.2 Classification. The knitted cloth shall be of the following types as specified (see 6.2):

Type I - Polypropylene, Brown 436
Type II - Polyester, Camouflage Green 483

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1 1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging of Synthetic Fiber Fabrics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. U S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

Method 135 - Dimensional Changes in Automatic Home Laundering of Durable Press Woven or Knit Fabrics

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O Box 12215, Research Triangle Park, NC 27709)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and appearance and shall, unless otherwise indicated (see 3.4.2), be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarn for type I cloth. The yarn shall be texturized or nontexturized multifilament polypropylene for both the backing and terry-loop face.

3.2.2 Yarn for type II cloth. The terry-loop yarn shall be 150/50 flat polyester. The ground yarn shall be 150/68 texturized polyester.

3.3 Knit. The cloth shall be a circular or warp knit terry loop cloth.

3.4 Color. The finished type I cloth shall be Brown 436 obtained by the use of melt-spun solution dyed fibers. The color of the finished type II cloth shall be dyed Camouflage Green 483 (see 6.3.1).

3.4.1 Matching. The color and appearance of the dyed and finished cloth shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the sample under incandescent lamplight at 2300 ± 200 K.

3.4.2 Colorfastness. The dyed and finished cloth shall show fastness to laundering and perspiration equal to or better than the standard sample or equal to or better than the rating of "good" when tested as specified in 4.2.3. The cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating not lower than 3.5 when tested as specified in 4.2.3.

3.5 Spectral reflectance. The dyed and finished type I and type II cloths shall conform to the reflectance values specified in table I when tested as specified in 4.2.3.

TABLE I. Spectral reflectance requirements

Wavelength Nanometers (nm)	Type I Reflectance (%)		Type II Reflectance (%)	
	Max.	Min.	Max.	Min.
600	18	12	10	3
620	18	12	10	3
640	19	12	10	3
660	19	13	11	3

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TABLE I. Spectral reflectance requirements (cont'd)

Wavelength Nanometers (nm)	Type I Reflectance (%) Max.	Min.	Type II Reflectance (%) Max.	Min.
680	20	13	13	3
700	20	13	28	4
720	21	13	40	5
740	21	13	52	7
760	22	14	60	11
780	22	14	64	17
800	23	14	70	24
820	23	15	80	32
840	24	15	86	37
860	24	15	88	40

3.6 Physical requirements. The finished cloth shall conform to the requirements specified in table II when tested as specified in 4.2.3.

TABLE II. Physical requirements

Characteristic	Type I	Type II
Weight, oz/sq/yd	5.9 (min)	6.0-6.5 (min-max)
Yarns per inch (min):		
Wales	24	21
Courses	22	31
Thickness, inch	0.065 (min)	0.062-0.072 (min-max)
Bursting strength, lbs (min)	80	100
Moisture absorption, sec. (max)		
Initial	-	5
After 15 launderings	-	30

3.6.1 Width. The minimum width of the finished cloth shall be 56 inches exclusive of any resin applied to the edges to prevent edge curling. The width of tubular circular knit cloth shall be obtained by slitting and finishing in such a manner as to prevent edge curling.

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3.7 Finish The cloth shall be scoured with detergent to remove impurities and no materials having a tendency to cause latent tendering or yellowing of the cloth shall be used in any of the dyeing or finishing operations. The type I cloth shall be brushed on the terry-loop face side. The terry-loop face of the type II cloth shall be napped and sheared to the finished thickness specified in 3.6 and shall exhibit no loose fiber. Both types shall be heat set to meet the requirements of this document.

3.7 1 Moisture absorption finish for type II cloth. If a finish is needed to meet the moisture absorption requirements specified for type II cloth in 3.6, the cloth shall be treated with an approved durable hydrophillic treatment (see 6.6)

3.8 Dimensional stability. The finished cloth shall not shrink or elongate more than 3.0 percent in either the wale or course direction when tested as specified in 4.2.3.

3 9 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.2.3.

3.10 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in one continuous piece of not less than 60 yards in length and shall be put up in rolls, full width, as specified in 5.1.

3.11 Fiber identification. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Textile Fiber Products Identification Act

3 12 Workmanship. The finished cloth shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4 1 2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4 2 Quality conformance inspection Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

4 2 1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4 2.2 End item examination.

4 2.2.1 Yard-by-yard examination. The cloth shall be examined at normal inspection distance (3 feet) for the defects listed below. All defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each yard (lengthwise) or fraction thereof in which it occurs. The lot size shall be expressed in yards. The sample unit shall be 1 linear yard. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0. The number of rolls from which the sample yardage is to be selected shall be in accordance with table III. The sample yardage shall be apportioned equally among the selected rolls.

Defects

Hole, cut, or tear
Drop stitch
Abrasion mark
Loose, broken, or missing yarn
Slub or slug
Float
Uneven, ridgy, or cockled cloth
Thin place
Spot, stain, or streak
Foreign matter
Barre, (greater than standard sample)

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4.2.2.2 Overall examination. The cloth shall be examined for the defects listed below. Each defect listed shall be counted not more than once in each roll examined. The sample size shall be the applicable number of rolls indicated in table III Each roll in the sample shall be rejected if the total number of defects in the sample exceeds the applicable acceptance number specified in table III

Defects

Off shade
Objectionable odor
Overall uncleanness
Uneven knitting throughout the roll
Edges frayed, ravelled, curled, rolled, folded, scalloped, or uneven continuously or intermittently throughout the roll
Overall width less than specified
Not labeled in accordance with the Textile Fiber Products Identification Act

TABLE III. Sample size

Lot size in yards	Sample size in rolls	Acceptance number
Up to 1,300 1/	3	0
1,301 up to and including 3,200	5	0
3,201 up to and including 8,000	7	0
8,001 up to and including 22,000	10	1
22,001 and over	15	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined

4.2.2.3 Length examination. Each roll of cloth used in the yard-by-yard examination shall be examined for the defects listed below. If the total number of defects in the sample rolls exceeds the applicable acceptance number specified in table III or if the total of the actual lengths of the sample rolls is less than the total of the lengths marked on the roll tickets, the lot shall be rejected.

Defects

Any gross length less than the specified minimum length.
Any gross length more than 2 yards less than gross length marked on roll ticket.

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4.2.2.4 Shade and appearance examination. A sample from each roll in the lot shall be examined visually for shade and appearance (see 3.1 and 3.4.1). The sample shall be an area not less than 5 inches by 8 inches. If any sample fails to meet the requirements for shade and appearance, the roll from which that sample was selected shall be rejected.

4 2 3 End item testing The cloth shall be tested for the characteristics listed in table IV The methods of testing specified in FED-STD-191 wherever applicable and as listed in table IV shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3-1/2 continuous yards full width of the finished cloth The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (no. of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. End item tests

Characteristic	Requirement paragraph	Test method
Yarn	3.2.1, 3.2.2	1/
Knit	3.3	1/
Colorfastness to:		
Laundering (after 3 cycles)	3 4.2	5614 2/
Perspiration	3.4.2	5680
Crocking	3 4.2	5651
Spectral reflectance	3 5	4.3.1
Weight	3 6	5041
Wales and courses per inch	3.6	5070
Thickness	3.6	5030
Bursting strength	3.6	5120
Moisture absorption (type II):		
Initial	3.6	4.3.2.1
After 15 launderings	3.6	and 4.3.2.2
Finish	3.7	1/
Dimensional stability	3.8	3/
pH	3.9	2811

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1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ Omit pressing.

3/ AATCC Method 135, Dimensional Changes in Automatic Home Laundering of Durable Press Woven or Knit Fabrics - Machine Wash Condition II (120° ± 5°F) Drying Procedure B (Tumble) except that the drying temperature shall not exceed 100°F.

4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1133

4.3 Methods of inspection.

4.3.1 Spectral reflectance test. Spectral reflectance data shall be obtair from 600-860 nm relative to a barium sulfate standard, the preferred white reference standard. Other reference white standards may be used provided they are calibrated to an absolute white; i.e., Halon, magnesium oxide, or vitrolit tile. The spectral band width at 860 nm shall be less than 26 nm. Reflectanc measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65 in the visible spectrum. The specular component of reflection shall be included in t measurement. Specimens shall be measured as a single layer of fabric backed t three layers of the same fabric. Readings will be taken on a minimum of two different areas and the data averaged. The specimen shall be viewed at an ang no greater than 10 degrees from normal. Photometric accuracy of the spectro-photometer shall be within 1 percent; and the wavelength accuracy shall be within 2 nm The standard aperture size used in the color measurement device shall be 1.0 to 1.25 inches in diameter. Failure of the spectral reflectance requirements will have occurred when spectral reflectance limits are exceeded four or more specified wavelengths.

4 3 2 Moisture absorption test.

4.3.2.1 Moisture absorption test, initial (type II). For each sample unit, 10 specimens shall be cut from an initial unlaundered sample. The minimum specimen size shall be 20 square inches Lay each specimen relaxed on a flat horizontal surface with the fabric napped surface facing up. Fill a straight medicine dropper with distilled water at 70° ± 2°F such that 1 drop equals approximately 0.05 mL. Hold dropper tip approximately 0.4 to 0.5 inches above the fabric surface at approximately a 45 degree angle and freely release one 0.05 mL drop onto the center area of the fabric surface. Once the water drop has been deposited onto the fabric, the specimen shall not be moved or agitate until the test is complete. Record the time for the water drop to be absorbed by the fabric. The drop is considered absorbed when it demonstrates a flat, dull

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appearance such that there is no light reflected off the droplet. Repeat this procedure for each specimen Average the ten results, and report to the nearest 1 second

4.3.2 2 Moisture absorption test, laundered (type II). For each sample unit, 10 specimens shall be cut from a sample laundered fifteen times in accordance with AATCC Method 135 using Machine Wash Condition II (120° ± 5°F) Drying Procedure B (Tumble) except that the drying temperature shall not exceed 100°F and the last two laundering cycles of the 15 total cycles shall be run without detergent. The ten specimens shall then be tested for moisture absorption in accordance with the procedure in 4.3.2.1 The results shall be averaged and reported to nearest 1 second.

5 PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1133.

5 2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2 1 Levels A, B, and Commercial. The cloth shall be packed in accordance with the applicable requirement of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1133.

6. NOTES

6 1 Intended use. The type I cloth is intended for use in cold weather undershirts and drawers while type II cloth is intended for use as the lining material in cold weather balaclavas.

6 2 Ordering data. Acquisition documents should specify the following:

a Title, number, and date of this document.
b. Cloth type (see 1.2).
c. Length and put-up if other than specified (see 3.10).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2)

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6.3 Standard sample. For access to standard sample, address the contrac activity issuing the invitation for bids

6.3.1 Dye formulation, type II. A suggested but not mandatory dye formulation for Camouflage Green 483 is as follows·

Disperse Yellow C.I. 1108
Disperse Blue C.I. 60
Disperse Red C.I. 167

6.4 White standards Barium sulfate of suitable quality for use as a wh standard is available from Eastman Kodak Company. Tiles are available from instrument manufacturer (see 4.3.1).

6.5 Spectrophotometers. Suitable spectrophotometers for measuring spect reflectance in the visible/near-infrared as the Diano Hardy, Diano Match-Sc Hunter D545P-IR, Hunter VIS/NIR Spectrocolorimeter, and Macbeth 1500 with I option

6.6 Approved type II finish. Approval of hydrophillic finish for type I cloth (see 3.7.1) is the responsibility of the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014. Approval is b on extensive evaluations, including those for toxicity, which are not set f in the document. Because of the time necessary to conduct full evaluation, those hydrophillic finishes already approved and so listed in the invitatio bids or request for proposal shall be considered acceptable for the related procurement.

6.6 1 Submittal for approval Request for finish approval should be sub mitted to the U S. Army Natick Research, Development, and Engineering Cente Natick, MA 01760-5014. It should be accompanied by approximately 5 linear yards of the type II cloth with the proposed hydrophillic finish and otherw conforming to all requirements for the type II cloth under this document together with applicable test data. Information is to be furnished regardi the commercial designation and chemical description of the hydrophillic fin (Approved list will indicate only the commercial designation). Chemical identification with products already approved should be made, if applicable any available data on skin irritancy or toxicity should be submitted. If applicable, a statement should be included that; as used, the finish is wel cured and that no materials with known high toxicity have been used prior t cure.

6.7 Subject term (key word) listing.

Cloth, knitted terry
Polyester
Polypropylene
Synthetic Fiber

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6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

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Navy - NU

Review activities:

Army - MD
Navy - MC
DLA - CT

Preparing activity:

Army - GL

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44161A	Cloth, Knitted, Terry, Synthetic Fibers

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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